

Yuehao Hu

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Skills

Languages: C/C++, Go, Python, Java, C#, Assembly | Systems & Technologies: Linux, VxWorks, RDMA, SPDK, MPI, RocksDB
Tools: Git, Docker, Jenkins, Qt, Jira, SVN, Keil

Work Experience

Distributed Storage Development Engineer | Enmotech

Mar 2021 – Jan 2024

- Contributed to the early design and development of the Chunk Server in the zStorage **distributed storage engine**, focusing on high-performance systems components.
- Designed and implemented a **Multi-Raft** consensus module on a fully **asynchronous SPDK** framework, supporting replicated data management and failure recovery.
- Contributed to I/O performance optimization, helping the early version achieve **300K+ IOPS** per node with average latency as low as **0.8 ms**; optimized the data path using **zero-copy** techniques and implemented a **shared-memory-based fio benchmark module**.
- Developed management plane features utilizing **Go, Gin, and RESTful APIs** to facilitate capacity balancing, QoS policies, and scalable, containerized multi-architecture deployments.

Software Engineer | Chengdu Xingduowei Technology

Jan 2020 – Feb 2021

- Developed firmware for **STM32**-based industrial CNC motion controllers.
- Led the design and implementation of **motion planning algorithms**, including linear and S-curve acceleration profiles and velocity lookahead optimization.
- Redesigned the CAD file processing algorithm in the host-side control software (**C++**, **Qt**), reducing processing time for large files by up to 300x.

Software Engineer | Alpha Networks

Mar 2017 - Jun 2019

- Developed and maintained enterprise Ethernet switch firmware on the **VxWorks** real-time operating system (**RTOS**).
- Developed **hardware drivers** for low-level components, including the PoE subsystem, PHY, power-management, and Flash storage.
- Designed and implemented **D-Link PD-Alive** functionality using **ICMP/ICMPv6** probing and **ACL-based** traffic monitoring to detect and maintain the operation of PoE-powered devices.

Publications

FARLock: Asymmetric RDMA Locking Made Fair | [Paper](#) | [Presentation](#)

- Designed and implemented **FARLock**, a high-performance fair lock based on **RDMA**, maintaining high throughput while significantly reducing tail latency, applicable to time-sensitive applications in cloud data centers.
- Developed a scalable **C++** benchmark framework for RDMA locks and automated performance evaluation with **Python**.
- Integrated **FARLock** into **Sherman**, an **RDMA-based B+ Tree** indexing system, achieving up to 14× tail-latency improvement across different workloads.

Education

MSc | Computer Science | GPA: 4.20/4.33 | Simon Fraser University

2024 – 2026

BEng | Internet of Things (IoT) Engineering | GPA: 85/100 (Top 5%) | Chengdu University of Technology

2011 – 2015